Probability

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I. INTRODUCTION

II. RANDOM EVENTS

Given an *elementary event* with a countable set of discrete random outcomes, $E_1, E_2, \ldots, E_n, \ldots$, there is associated with each possible outcome E_k a number called a *probability*, p_k, which can lie between 0 and 1, $0 \leq p_k \leq 1$.

If the kth outcome never occurs, $p_k = 0$; if it is sure to occur, $p_k = 1$. Conversely, if $p_k = 0$, we say that the event almost surely does not occur; and if $p_k = 1$, the event almost surely occurs. Another notation for the probability of event E_k is $P\{E_k\} = p_k$.

The following are some simple properties of the probability of events:

1. $P\{E_i \text{ and/or } E_j\} \leq p_i + p_j$.

2. E_i and E_j are said to be *mutually exclusive* events if and only if the occurrence of E_i implies that E_j does not occur, $E_i \implies \bar{E}_j$ (not E_j), and vice versa. If E_i and E_j are mutually exclusive, $P\{E_i \text{ and } E_j\} = 0, P\{E_i \text{ or } E_j\} = p_i + p_j$. 3. A whole class of events can be mutually exclusive for all i and j. When the class is *exhaustive*, that is, all possible events have been enumerated, $P\{\text{some } E_i\} = \sum_i p_i = 1$.

   In the following, we consider a compound experiment consisting of just two elementary events. For clarity, we imagine the first to have outcomes $\{E_i\}$ with probability p_{1i} and the second to have outcomes $\{F_j\}$ and probabilities p_{2j}, respectively. Each of p_{1j} and p_{2j} obeys statements 1, 2, and 3 above. An outcome of such a composite event is a pair (E_i, F_j).

4. The probability of the specific outcome (E_i, F_j) is p_{ij}, called the *joint probability* for E_i and F_j.

5. $p_{ij} = p_{1i} \cdot p_{2j}$, if and only if events E_i and F_j are *independent*.

6. Suppose E_i and F_j are not independent; then the joint probability can be written as

$$p_{ij} = \left(\sum_k p_{ik}\right)\left(\frac{p_{ij}}{\sum_k p_{ik}}\right) = p(i)\left(\frac{p_{ij}}{\sum_k p_{ik}}\right), \tag{2.1}$$

   $p(i)$ defines a new number called the *marginal probability* for event E_i, that is, the probability that E_i does, in fact, occur, whatever the second event may be. Therefore,

$$\sum_i p(i) = \sum_i \sum_k p_{ik} = 1 \tag{2.2}$$

   The marginal distribution for the second event, F_j, can similarly be written as $p(j) = \sum_k p_{kj}$.

7. The second factor of Eq. (2.1) is the *conditional probability*

$$p(j|i) = \frac{p_{ij}}{\sum_k p_{ik}} \tag{2.3}$$

   and is the probability for event F_j occurring, given that event E_i has occurred. The probability for some F_j should be 1, and indeed

$$\sum_j p(j|i) = \sum_j \frac{p_{ij}}{\sum_k p_{ik}} = 1 \quad \text{for every } i. \tag{2.4}$$

All joint probabilities can be factored into a marginal distribution and a conditional probability. This scheme can be generalized to treat the joint occurrence of three or more elementary events.

III. RANDOM VARIABLES

In the following discussion we shall assume that for every elementary outcome E_i, there is an associated real number x_i. A random selection, X, of one of the possible values $x_1, x_2, \ldots$ is called a random variable. The probability that the value x_i is chosen is given by $p_i = P\{X = x_i\}$. The *expectation* of this random variable X, that is, the stochastic mean value, is defined as

$$E(X) \equiv \sum_i P\{X = x_i\} x_i = \sum_i p_i x_i = \mu. \tag{3.1}$$

μ is called the *expected* or *mean value*. It is common in physics to write this as $\langle X \rangle$, and we shall often use that notation.

Consider some real-valued function $g(x_i) = g_i$, where the x_i correspond to a countable set of elementary events with probabilities p_i. If X is a random variable, then $g(X)$ is also a random variable. The expectation of $g(X)$ is defined as

$$E(g(X)) = \langle g(X) \rangle = \sum_i p_i g(x_i). \tag{3.2}$$

From the definition of the expected value of a function, we have the property that

$$\langle \text{constant} \rangle = \text{constant}, \tag{3.3}$$

and that for any constants λ_1, λ_2 and two functions g_1, g_2,

$$\langle \lambda_1 g_1(X) + \lambda_2 g_2(X) \rangle = \lambda_1 \langle g_1(X) \rangle + \lambda_2 \langle g_2(X) \rangle. \tag{3.4}$$

An important application of expected values is to the powers of X. The nth moment of X is defined as the expectation of the nth power of X,

$$\langle X^n \rangle \equiv \sum_i p_i x_i^n, \tag{3.5}$$

The *central moments* of X are given by

$$\langle (X - \mu)^n \rangle \equiv \sum_i p_i (x_i - \langle X \rangle)^n. \tag{3.6}$$

The second central moment has particular significance,

$$\langle (X - \mu)^2 \rangle = \langle (X - \langle X \rangle)^2 \rangle = \sum_i p_i (x_i - \langle X \rangle)^2 = \sum_i p_1 x_i^2 - \langle X \rangle^2 = \langle X^2 \rangle - \langle X \rangle^2, \tag{3.7}$$

and is called the *variance* of X or $\text{var}\{X\}$. The square root of the variance is a measure of the dispersion of the random variable. It is referred to as the *standard deviation* and sometimes the *standard error*. The variance of a function of the random variable, $g(X)$, can be determined as

$$\text{var}\{g(X)\} = \langle (g(X) - \langle g(X) \rangle)^2 \rangle = \langle g^2(X) \rangle - \langle g(X) \rangle^2. \tag{3.8}$$

Consider two random variables, X and Y. They are are not in general independent. Two random variables are said to be independent if they derive from independent events. As we have seen in Eq. (3.4), the expectation of a linear combination is the linear combination of the expectations. This result does not require that X and Y be independent. The effect of statistical dependence will be seen in the variance of a linear combination of the two variables,

$$\text{var}\{\lambda_1 X + \lambda_2 Y\} = \lambda_1^2 \text{var}\{X\} + \lambda_2^2 \text{var}\{Y\} + 2[\lambda_1 \lambda_2 \langle XY \rangle - \lambda_1 \lambda_2 \langle X \rangle \langle Y \rangle]. \tag{3.9}$$

The expectation of the product is

$$\langle XY \rangle = \sum_{i,j} p_{ij} x_i y_j. \tag{3.10}$$

If X and Y are independent, $p_{ij} = p_{1i} \cdot p_{2j}$ and

$$\langle XY \rangle = \sum_i p_{1i} x_i \sum_j p_{2j} y_j = \langle X \rangle \langle Y \rangle. \tag{3.11}$$

Then Eq. (3.9) becomes simply

$$\mathrm{var}\{\lambda_1 X + \lambda_2 Y\} = \lambda_1^2 \mathrm{var}\{X\} + \lambda_2^2 \mathrm{var}\{Y\}. \tag{3.12}$$

When X and Y are not necessarily independent, we introduce a new quantity, the *covariance*, which is a measure of the degree of independence of the two random variables X and Y:

$$\mathrm{cov}\{X, Y\} = \langle XY \rangle - \langle X \rangle \langle Y \rangle. \tag{3.13}$$

The covariance equals 0 when X and Y are independent and $\mathrm{cov}\{X, X\} = \mathrm{var}\{X\}$.

Note that zero covariance does not by itself imply independence of the random variables. The following simple example illustrates that even functional dependence can still yield a zero covariance. Let X be a random variable that may be -1, 0, or 1 with equal probabilities, and define $Y = X^2$. Obviously, $\langle X \rangle = 0, \langle XY \rangle = \langle X^3 \rangle = 0$ so $\mathrm{cov}\{XY\} = 0$.

The covariance can have either a positive or negative value. Another quantity derived from the covariance is the correlation coefficient,

$$\rho(X, Y) = \frac{\mathrm{cov}\{X, Y\}}{\sqrt{\mathrm{var}\{X\}\mathrm{var}\{Y\}}}, \tag{3.14}$$

so that $-1 \leq \rho(X, Y) \leq 1$. Since the covariance can be positive or negative, the variance of a linear combination of two dependent random variables can be greater or less than the variance if the variables were independent.

A. The Binomial Distribution

Consider two events E_0 and E_1 that are mutually exclusive and exhaustive:

$$P\{E_1\} = p, \quad x_1 = 1, \tag{3.15}$$

$$P\{E_0\} = 1 - p, \quad x_0 = 0. \tag{3.16}$$

Let X_i be the random variable that is the ith outcome of a series of N such events. The expectations of X_i and its square become

$$E(X_i) = p \cdot 1 + (1 - p) \cdot 0 = p, \tag{3.17}$$

$$E(X_i^2) = p, \tag{3.18}$$

and the variance is then

$$\mathrm{var}\{X_i\} = p - p^2 = p(1 - p). \tag{3.19}$$

Each outcome is either 0 or 1, and we set X to be the sum of the N outcomes $\{X_i\}$,

$$X = \sum_{i=1}^{N} X_i. \tag{3.20}$$

The probability that $X = n$ is the probability that n of the X_i were 1 and $N - n$ were 0. That is,

$$P\{X = n\} = \binom{N}{n} p^n (1 - p)^{N-n} \quad n = 0, 1, \ldots, N. \tag{3.21}$$

This is the *binomial distribution*. $\binom{N}{n} = N!/n!(N - n)!$ is the binomial coefficient, which counts the number of different ways in which the n E_1 and the $(N - n)$ E_0 may occur. The expected value of X is

$$\langle X \rangle = \sum_{n=0}^{N} n \binom{N}{n} p^n (1 - p)^{N-n} = Np. \tag{3.22}$$

This may be verified by direct computation of the sum or by noting that the expected value of X is the sum of the expected value of all the X_i. The variance of X is easily determined; since the X_i are independent, the result in Eq. (3.12) may be employed and

$$\langle (X - Np)^2 \rangle = \sum_{i=1}^{N} \mathrm{var}\{X_i\} = \sum_{i=1}^{N} p(1 - p) = Np(1 - p). \tag{3.23}$$

B. The Geometric Distribution

Suppose we carry out a certain experiment repeatedly and independently, where there are only two outcomes: failure or success. If the outcome is a failure, the experiment will be repeated; otherwise, we stop the procedure. Let the random variable X of interest be the number of times we repeat the experiment until the first success. Let q be the probability of failure in one experiment and $p = 1 - q$ be the probability of success. Then

$$P\{X = n\} = q^{n-1}p, \quad n = 1, 2, \ldots. \tag{3.24}$$

The average number of experiments that will be carried out is

$$\langle X \rangle = \sum_{n=1}^{\infty} nq^{n-1}p = \frac{p}{(1-q)^2} = \frac{1}{p}. \tag{3.25}$$

The variance of X can be calculated as

$$\mathrm{var}\{X\} = \langle X^2 \rangle - \langle X \rangle^2 = \left(\frac{2}{p^2} - \frac{1}{p}\right) - \frac{1}{p^2}. \tag{3.26}$$

As an example, in particle transport problems, the number of collisions one particle makes follows this distribution if the medium is infinite, homogeneous, and if the relative probability of different outcomes is constant.

C. The Poisson Distribution

A random variable X is said to follow a *Poisson distribution* when

$$P\{X = n\} = \frac{\lambda^n}{n!}e^{-\lambda}, \quad n = 0, 1, \ldots, \tag{3.27}$$

where λ is a parameter of the distribution. It is easy to show that

$$\langle X \rangle = \lambda, \tag{3.28}$$

$$\mathrm{var}\{X\} = \lambda. \tag{3.29}$$

This distribution is fundamental in the theory of probability and stochastic processes. It is of great use in applications such as research into queuing service systems and similar discrete event problems.

IV. CONTINUOUS RANDOM VARIABLES

V. EXPECTATIONS OF CONTINUOUS RANDOM VARIABLES

VI. BIVARIATE CONTINUOUS RANDOM DISTRIBUTIONS

VII. SUM OF RANDOM VARIABLES

VIII. DISTRIBUTION OF THE MEAN OF A RANDOM VARIABLE

IX. DISTRIBUTION OF THE SUM OF INDEPENDENT RANDOM VARIABLE

The *central limit theorem* has a proof using characteristic functions.

Assume $\{X_1, \ldots, X_n, \ldots\}$ are independent and identically distributed random variables, each with mean μ and finite variance σ^2. The sum $X_1 + \cdots + X_n$ has mean $n\mu$ and variance $n\sigma^2$. Consider the random variable

$$Z_n = \frac{X_1 + \cdots + X_n - n\mu}{\sqrt{n\sigma^2}} = \sum_{i=1}^{n} \frac{X_i - \mu}{\sqrt{n\sigma^2}} = \sum_{i=1}^{n} \frac{1}{\sqrt{n}} Y_i, \tag{9.1}$$

where in the last step we defined the new random variables $Y_i = \frac{X_i - \mu}{\sigma}$, each with zero mean and unit variance $\mathrm{var}(Y) = 1$). The characteristic function of Z_n is given by

$$\begin{aligned} \varphi_{Z_n}(t) = \varphi_{\sum_{i=1}^n \frac{1}{\sqrt{n}} Y_i}(t) &= \varphi_{Y_1}\left(\frac{t}{\sqrt{n}}\right) \varphi_{Y_2}\left(\frac{t}{\sqrt{n}}\right) \cdots \varphi_{Y_n}\left(\frac{t}{\sqrt{n}}\right) \\ &= \left[\varphi_{Y_1}\left(\frac{t}{\sqrt{n}}\right)\right]^n, \end{aligned} \tag{9.2}$$

where in the last step we used the fact that all of the Y_i are identically distributed. The characteristic function of Y_1 is, by Taylor theorem,

$$\varphi_{Y_1}\left(\frac{t}{\sqrt{n}}\right) = 1 - \frac{t^2}{2n} + o\left(\frac{t^2}{n}\right), \quad \left(\frac{t}{\sqrt{n}}\right) \to 0 \tag{9.3}$$

where $o(t^2/n)$ is "little o notation" for some function of t that goes to zero more rapidly than t^2/n. By the limit of the exponential function $e^x = \lim_{n\to\infty}\left(1 + \frac{x}{n}\right)^n$, the characteristic function of Z_n equals

$$\varphi_{Z_n}(t) = \left(1 - \frac{t^2}{2n} + o\left(\frac{t^2}{n}\right)\right)^n \to e^{-\frac{1}{2}t^2}, \quad n \to \infty. \tag{9.4}$$

All of the higher order terms vanish in the limit $n \to \infty$. The right hand side equals the characteristic function of a standard normal distribution $\mathscr{N}(0,1)$, which implies through *Lévy continuity theorem* [1] that the distribution of Z_n will approach $\mathscr{N}(0,1)$ as $n \to \infty$. Therefore, the sample average

$$\bar{X}_n = \frac{X_1 + \cdots + X_n}{n} \tag{9.5}$$

is such that

$$\frac{\sqrt{n}}{\sigma}\left(\bar{X}_n - \mu\right) = Z_n \tag{9.6}$$

converges to the normal distribution $\mathscr{N}(0,1)$, from which the central limit theorem follows.

X. MONTE CARLO INTEGRATION

XI. MARKOV PROCESSES

The playground for random variables are *stochastic processes* of which the *Markov process* plays a privileged role since it allows a more complete theory.

A. Stochastic Processes

We are now embarking on an essential aspect of our description, namely, how stochastic aspects of noise $\eta(t)$ are properly accounted for. Obviously, a particular realization of the time-dependent process $\eta(t)$ does not provide much information. Rather, one needs to consider the probability of observing a certain sequence of noise values $\eta_1, \eta_2, \ldots$ at times $t_1, t_2, \ldots$. The essential information is entailed in the conditional probabilities (2.3)

$$p(\eta_1, t_1; \eta_2, t_2; \ldots | \eta_0, t_0; \eta_{-1}, t_{-1}; \ldots), \tag{11.1}$$

when the process is assumed to generate noise at fixed times t_i, $t_i < t_j$ for $i < j$. Here $p(|)$ is the probability that the random variable $\eta(t)$ assumes the values $\eta_1, \eta_2, \ldots$ at times $t_1, t_2, \ldots$, if it had previously assumed the values $\eta_0, \eta_{-1}, \ldots$ at times $t_0, t_{-1}, \ldots$.

B. The Markov case

An important class of random processes are the so-called *Markov processes* for which the conditional probabilities depend only on η_0 and t_0 and not on earlier occurrences of noise values. In this case holds

$$p(\eta_1, t_1; \eta_2, t_2; \ldots | \eta_0, t_0; \eta_{-1}, t_{-1}; \ldots) = p(\eta_1, t_1; \eta_2, t_2; \ldots | \eta_0, t_0). \tag{11.2}$$

This property allows one to factorize $p(|)$ into a sequence of consecutive conditional probabilities.

$$\begin{aligned} p(\eta_1, t_1; \eta_2, t_2; \ldots | \eta_0, t_0; \eta_{-1}, t_{-1}; \ldots) &= p(\eta_2, t_2; \eta_3, t_3; \ldots | \eta_1, t_1) p(\eta_1, t_1 | \eta_0, t_0) \\ &= p(\eta_3, t_3; \eta_4, t_4; \ldots | \eta_2, t_2) p(\eta_2, t_2 | \eta_1, t_1) p(\eta_1, t_1 | \eta_0, t_0) \\ &\vdots \end{aligned}$$

The unconditional probability for the realization of $\eta_1, \eta_2, \ldots$ at times $t_1, t_2, \ldots$ is

$$p(\eta_1, t_1; \eta_2, t_2; \ldots) = \sum_{\eta_0} p(\eta_0, t_0) p(\eta_1, t_1 | \eta_0, t_0) p(\eta_2, t_2 | \eta_1, t_1) \cdots , \tag{11.3}$$

where $p(\eta_0, t_0)$ is the unconditional probability for the appearance of η_0 at time t_0. One can conclude from Eq. (11.3) that a knowledge of $p(\eta_0, t_0)$ and $p(\eta_i, t_i | \eta_{i-1}, t_{i-1})$ is sufficient for a complete characterization of a Markov process.

Before we proceed with three important examples of Markov processes we will take a short detour and give a quick introduction on mathematical tools that will be useful in handling probability distributions like $p(\eta_0, t_0)$ and $p(\eta_i, t_i | \eta_{i-1}, t_{i-1})$.

C. Characteristics of Probability Distributions

For simplicity we now deal with a one-dimensional random variable η with values on the complete real axis. In probability theory the Fourier transform $G(s, t)$ of $p(\eta, t)$ is referred to as the *characteristic function* of $p(\eta, t)$.

$$G(s, t) = \int_{-\infty}^{\infty} d\bar{\eta}\, p(\bar{\eta}, t) e^{i\bar{\eta} s} \tag{11.4}$$

Since the Fourier transform can be inverted to yield $p(\eta, t)$

$$p(\eta, t) = \int_{-\infty}^{\infty} \frac{ds}{2\pi}\, G(s, t) e^{-is\eta}, \tag{11.5}$$

$G(s, t)$ contains all information on $p(\eta, t)$.

The characteristic function can be interpreted as an average of $e^{is\eta(t)}$, and denoted by

$$G(s, t) = \left\langle e^{is\eta(t)} \right\rangle . \tag{11.6}$$

Eq. (11.6) prompts one to consider the Taylor expansion of (11.6) for (is) around 0:

$$G(s, t) = \sum_{n=0}^{\infty} \langle \eta^n(t) \rangle \frac{(is)^n}{n!}, \tag{11.7}$$

where

$$\langle \eta^n(t) \rangle = \int d\eta\, \eta^n p(\eta, t). \tag{11.8}$$

are the so-called *moments* of $p(\eta, t)$. One can conclude from (11.5), (11.7), and (11.8) that the moments $\langle \eta^n(t) \rangle$ completely characterize $p(\eta, t)$.

The moments $\langle \eta^n(t) \rangle$ can be gathered in a statistical analysis as averages of powers of the stochastic variable $\eta(t)$. Obviously, it is of interest to employ averages which characterize a distribution $p(\eta, t)$ as succinctly as possible, i.e., through the smallest number of averages. Unfortunately moments $\langle \eta^n(t) \rangle$ of all orders of n contain significant information about $p(\eta, t)$. There is another, similar, but more useful scheme to describe probability distributions

$p(\eta, t)$; the *cumulants* $\langle\langle \eta^n(t) \rangle\rangle$. As moments are generated by the characteristic function $G(s,t)$, cumulants are generated by the logarithm of the characteristic function $\ln[G(s,t)]$

$$\ln[G(s,t)] = \sum_{n=1}^{\infty} \langle\langle \eta^n(t) \rangle\rangle \frac{(is)^n}{n!}. \tag{11.9}$$

Cumulants can be expressed in terms of $\langle \eta^n(t) \rangle$ by taking the logarithm of Eq. (11.7) and comparing the result with (11.9). The first three cumulants are

$$\langle\langle \eta^1(t) \rangle\rangle = \langle \eta^1(t) \rangle, \tag{11.10}$$

$$\langle\langle \eta^2(t) \rangle\rangle = \langle \eta^2(t) \rangle - \langle \eta^1(t) \rangle^2, \tag{11.11}$$

$$\langle\langle \eta^3(t) \rangle\rangle = \langle \eta^3(t) \rangle - 3 \langle \eta^2(t) \rangle \langle \eta^1(t) \rangle + 2 \langle \eta^1(t) \rangle^3. \tag{11.12}$$

These expressions reveal that the first cumulant is equal to the average of the stochastic variable and the second cumulant is equal to the variance (3.8). The higher orders of cumulants contain less information about $p(\eta, t)$ than lower ones. In fact it can be shown, that in the frequently arising case of probabilities described by Gaussian distributions (the corresponding random processes are called Gaussian) all, but the first and second-order cumulants vanish. For non-Gaussian distributions, though, all cumulants are non-zero as stated in the theorem of Marcienkiewicz [2]). Nevertheless, cumulants give a more succinct description of $p(\eta, t)$ than moments do, dramatically so in case of Gaussian processes. This is not the only benefit as we will see considering scenarios with more than one random variable $\eta(t)$.

Cumulants are particularly useful if one has to consider the sum of statistically independent random values, for example the sum $\sigma = \eta_0 + \eta_1$. In this case one can easily prove that the characteristic function for the two variables η_0 and η_1 is the product of the two characteristic functions for each variable. So its logarithm is the sum of the logarithms of the two characteristic functions. Therefore a Taylor expansion shows that $\langle\langle \eta_0^{n_0} \eta_1^{n_1} \rangle\rangle = 0$ for all $n_0, n_1 \geq 1$. Thus $\langle\langle (\eta_0 + \eta_1)^n \rangle\rangle = \langle\langle \eta_0^n \rangle\rangle + \langle\langle \eta_1^n \rangle\rangle$. This result implies that cumulants of any order are simply added if one accumulates the corresponding statistically independent random variables, hence the name "cumulant". For an arbitrary number of statistically independent random variables η_j or even continuously many $\eta(t)$ one can write

$$\left\langle\left\langle \left(\sum_j \eta_j \right)^n \right\rangle\right\rangle = \sum_j \langle\langle \eta_j^n \rangle\rangle, \tag{11.13}$$

$$\left\langle\left\langle \left(\int dt\, \eta(t) \right)^n \right\rangle\right\rangle = \int dt\, \langle\langle \eta^n(t) \rangle\rangle, \tag{11.14}$$

properties which will be utilized below.

We will now furnish concrete, analytic expressions for the probabilities characterizing three important Markov processes. We begin with the so-called Wiener process.

D. Wiener Process

The *Wiener process*, described by $\omega(t)$ for $t > 0$, is characterized by the probability distributions

$$p(\omega_0, t_0) = \frac{1}{\sqrt{4\pi D t_0}} e^{-\omega_0^2/4Dt_0}, \tag{11.15}$$

$$p(\omega_1, t_1 | \omega_0, t_0) = \frac{1}{\sqrt{4\pi D \Delta t}} e^{-(\Delta\omega)^2/4D\Delta t}, \tag{11.16}$$

$$\Delta\omega = \omega_1 - \omega_0, \tag{11.17}$$

$$\Delta t = t_1 - t_0. \tag{11.18}$$

The probabilities are parameterized through the constant D, referred to as the diffusion constant, since the probability distributions $p(\omega_0, t_0)$ and $p(\omega_1, t_1 | \omega_0, t_0)$ are solutions of the diffusion equation (**??**) discussed extensively below. The Wiener process is homogeneous in time and space, which implies that the conditional transition probability $p(\omega_1, t_1 | \omega_0, t_0)$ depends only on the relative variables $\Delta\omega$ and Δt. Put differently, the probability $p(\Delta\omega, \Delta t)$ for an increment $\Delta\omega$ to occur is independent of the current state of the Wiener process $\omega(t)$. The probability $p(\Delta\omega, \Delta t) = p(\omega_0 + \Delta\omega, t_0 + \Delta t | \omega_0, t_0)$ is given by Eq. (11.16).

Characteristic Functions, Moments, Correlation Functions and Cumulants for the Wiener process

In case of the Wiener process simple expressions can be provided for the characteristic function, moments and cumulants. Combining (11.16) and (11.5) one obtains for the characteristic function

$$G(s,t) = e^{-Dts^2}. \tag{11.19}$$

A Taylor expansion allows one to identify the moments [1]

$$\langle \omega^n(t) \rangle = \begin{cases} 0 & \text{for odd } n, \\ (n-1)!!(2Dt)^{n/2} & \text{otherwise.} \end{cases} \tag{11.20}$$

The definition (11.9) and (11.19) leads to the expression for the cumulants

$$\langle\langle \omega^n(t) \rangle\rangle = \begin{cases} 2Dt & \text{for } n = 2, \\ 0 & \text{otherwise.} \end{cases} \tag{11.21}$$

For the two-dimensional characteristic functions one can derive, using (11.16) and (11.6)

$$G(s_0, t_0; s_1, t_1) = e^{-D(s_0^2 t_0 + s_1^2 t_1 + 2 s_0 s_1 \min(t_0, t_1))}, \tag{11.22}$$

From this follow the correlation functions

$$\langle \omega^{n_1}(t_1) \omega^{n_0}(t_0) \rangle = \begin{cases} 0 & \text{for odd } n_0 + n_1, \\ 2D\min(t_0, t_1) & \text{for } n_0 = n_1 = 1, \\ 12D^2 t_0 \min(t_0, t_1) & \text{for } n_0 = 1, n_1 = 3, \\ 4D^2[t_0 t_1 + 2\min^2(t_0, t_1)] & \text{for } n_0 = 2, n_1 = 2, \\ \dots \end{cases} \tag{11.23}$$

and, using the definition (11.9), the cumulants

$$\langle\langle \omega^{n_1}(t_1) \omega^{n_0}(t_0) \rangle\rangle = \begin{cases} 2D\min(t_0, t_1) & \text{for } n_0 = n_1 = 1, \\ 0 & \text{otherwise for } n_0, n_1 \neq 0. \end{cases} \tag{11.24}$$

The Wiener Process as the Continuum Limit of a Random Walk on a Lattice

The Wiener process is closely related to a random walk on a one-dimensional lattice with lattice constant a. A n-step walk on a lattice is performed in discrete times steps $t_j = j\tau$, with $j = 0, 1, 2, \dots, n$. The walk may start at an arbitrary lattice site x_0. One can choose this starting position as the origin of the coordinate system so that one can set $x_0 = 0$. The lattice sites are then located at $x_i = ia, i \in \mathbb{Z}$. At each time step the random walker moves with equal probability to the neighboring right or left lattice site. Thus, after the first step with $t = \tau$ one will find the random walker at $x = \pm a$, i.e. at site $x_{\pm 1}$ with probability $P(\pm a, \tau) = 1/2$. For a two-step walk the following paths are possible:

- path 1: two steps to the left,
- path 2: one step to the left and then one step to the right,
- path 3: one step to the right and then one step to the left,
- path 4: two steps to the right.

Each path has a probability of 1/4, a factor 1/2 for each step. Paths 2 and 3 both terminate at lattice site x_0. The probability to find a random walker after two step at position $x_0 = 0$ is therefore $P(0, 2\tau) = 1/2$. The probabilities for lattice sites $x_{\pm 2}$ reached via path 1 and 4 respectively are simply $P(\pm 2a, 2\tau) = 1/4$.

For an n-step walk one can proceed like this summing over all possible paths that terminate at a given lattice site x_i. Such a summation yields the probability $P(ia, n\tau)$. However, to do so effectively a more elegant mathematical description is appropriate. We denote a step to the right by an operator $\mathscr{R}$, and a step to the left by an operator $\mathscr{L}$.

---

[1] The double factorial $n!!$ for positive $n \in \mathbb{N}$ denotes the product $n(n-2)(n-4)\cdots 1$ for odd n and $n(n-2)(n-4)\cdots 2$ for even n.

Consequently a single step of a random walker is given by $\frac{1}{2}(\mathscr{L}+\mathscr{R})$, the factor $\frac{1}{2}$ denoting the probability for each direction. To obtain a n-step walk the above operator $\frac{1}{2}(\mathscr{L}+\mathscr{R})$ has to be iterated n times. For a two-step walk one gets $\frac{1}{4}(\mathscr{L}+\mathscr{R})\circ(\mathscr{L}+\mathscr{R})$. Expanding this expression results in $\frac{1}{4}(\mathscr{L}^2+\mathscr{L}\circ\mathscr{R}+\mathscr{R}\circ\mathscr{L}+\mathscr{R}^2)$. Since a step to the right and then to the left amounts to the same as a step first to the left and then to the right, it is safe to assume that $\mathscr{R}$ and $\mathscr{L}$ commute. Hence one can write $\frac{1}{4}\mathscr{L}^2+\frac{1}{2}\mathscr{L}\circ\mathscr{R}+\frac{1}{4}\mathscr{R}^2$. As the operator expression $\mathscr{L}^p\circ\mathscr{R}^q$ stands for p steps to the left and q steps to the right one can deduce that $\mathscr{L}^p\circ\mathscr{R}^q$ represents the lattice site x_{q-p}. The coefficients are the corresponding probabilities $P((q-p)a,(q+p)\tau)$. The algebraic approach above proves useful, since one can utilize the well known binomial formula

$$\left[\frac{1}{2}(\mathscr{L}+\mathscr{R})\right]^n=\left(\frac{1}{2}\right)^n\sum_{k=0}^{n}\binom{n}{k}\mathscr{L}^k\circ\mathscr{R}^{n-k}, \tag{11.25}$$

and obtains as coefficients of x_i the probabilities

$$P(ia,n\tau)=\frac{1}{2^n}\binom{n}{\frac{n+i}{2}}, \tag{11.26}$$

which can be expressed as

$$P(x,t)=\frac{1}{2^{t/\tau}}\binom{t/\tau}{\frac{t}{2\tau}+\frac{x}{2a}}. \tag{11.27}$$

The moments of the discrete probability distribution $P(x,t)$ are

$$\langle x^n(t)\rangle=\sum_{x=-\infty}^{\infty}x^nP(x,t)=\begin{cases}0 & \text{for odd } n,\\ a^2\frac{t}{\tau} & \text{for } n=2,\\ a^4\frac{t}{\tau}\left(3\frac{t}{\tau}-2\right) & \text{for } n=4,\\ a^6\frac{t}{\tau}\left(15\left(\frac{t}{\tau}\right)^2-30\frac{t}{\tau}+16\right) & \text{for } n=6,\\ \dots & .\end{cases} \tag{11.28}$$

We now want to demonstrate that in the continuum limit a random walk reproduces a Wiener process. For this purpose we show that the unconditional probability distributions of both processes match. We do not consider conditional probabilities $p(x_1,t_1|x_0,t_0)$ as they equal unconditional probabilities $p(x_1-x_0,t_1-t_0)$ in both cases; in a Wiener process as well as in a random walk. To turn the discrete probability distribution (11.27) into a continuous probability density distribution one considers adjacent bins centered on every lattice site that may be occupied by a random walker. Note, that only every second lattice site can be reached after a particular number of steps. Thus, these adjacent bins have a base length of $2a$ by which we have to divide $P(x,t)$ to obtain the probability density distribution $p(x,t)$ in these bins

$$p(x,t)\,dx=\frac{1}{2a}\frac{1}{2^{t/\tau}}\binom{t/\tau}{\frac{t}{2\tau}+\frac{x}{2a}}dx. \tag{11.29}$$

We then rescale the lattice constant a and the length τ of the time intervals to obtain a continuous description in time and space. However, τ and a need to be rescaled differently, since the spatial extension of the probability distribution $p(x,t)$, characterized by its standard deviation

$$\sqrt{\langle\langle x^2(t)\rangle\rangle}=a\sqrt{\frac{t}{\tau}} \tag{11.30}$$

is not proportional to t, but to $\sqrt{t}$. This is a profound fact and a common feature for all processes accumulating uncorrelated values of random variables in time. Thus, to conserve the temporal-spatial proportions of the Wiener process one rescales the time step τ by a factor ϵ and the lattice constant a by a factor $\sqrt{\epsilon}$

$$\tau\to\epsilon\tau \quad\text{and}\quad a\to\sqrt{\epsilon}a. \tag{11.31}$$

A continuous description of the binomial density distribution (11.29) is then approached by taking the limit $\epsilon\to 0$. In this limit the number of steps $n=\frac{t}{\epsilon\tau}$ in the random walk goes to infinity and one observes the following identity

derived in Appendix A

$$
\begin{aligned}
p(x,t)\,dx &= \frac{1}{2\epsilon a} 2^{-\frac{t}{\epsilon\tau}} \binom{\frac{t}{\epsilon\tau}}{\frac{t}{2\epsilon\tau}+\frac{x}{2\sqrt{\epsilon}a}}\,dx \\
&= \sqrt{\frac{n\tau}{4a^2t}}\,2^{-n}\binom{n}{\frac{n}{2}+\frac{x}{a}\sqrt{\frac{n\tau}{4t}}}\,dx \\
&= \sqrt{\frac{\tau}{2\pi a^2 t}}\,e^{-\frac{x^2\tau}{2a^2t}}\,dx\left(1+O\left(\frac{1}{n}\right)\right). \qquad (11.32)
\end{aligned}
$$

The fraction τ/a^2 is invariant under rescaling (11.31) and, hence, this quantity remains in the continuous description (11.32) of the probability density distribution $p(x,t)$. Comparing equations (11.32) and (11.16) one identifies $D = a^2/2\tau$. The relation between random step length a and time unit τ obviously determines the rate of diffusion embodied in the diffusion constant D: the larger the steps a and the more rapidly these are performed, i.e., the smaller τ, the quicker the diffusion process and the faster the broadening of the probability density distribution $p(x,t)$. According to (11.30) this broadening is then $\sqrt{2Dt}$ as expected for a diffusion process.

A Wiener Process can be Integrated, but not Differentiated

E. Ornstein-Uhlenbeck Process

Our second example for a Markov process is the *Ornstein-Uhlenbeck process*. The Ornstein-Uhlenbeck process, describing a random variable $v(t)$, is defined through the probabilities

$$
\begin{aligned}
p(v_0,t_0) &= \frac{1}{\sqrt{\pi\gamma\sigma^2}}e^{-v_0^2/\gamma\sigma^2}, && (11.33)\\
p(v_1,t_1|v_0,t_0) &= \frac{1}{\sqrt{\pi S}}e^{-\left(v_1-v_0e^{-\gamma\Delta t}\right)^2/S}, && (11.34)\\
S &= \gamma\sigma^2\left(1-e^{-2\gamma\Delta t}\right), && (11.35)\\
\Delta t &= |t_1-t_0|. && (11.36)
\end{aligned}
$$

The probabilities are characterized through two parameters σ and γ. Their significance will be explained further below. The process is homogeneous in time, since (11.34) depends solely on t, but is not homogeneous in v. Furthermore, the Ornstein-Uhlenbeck Process is stationary, i.e., $p(v_0,t_0)$ does not change in time.

Characteristic Functions, Moments, Correlation Functions and Cumulants for the Ornstein-Uhlenbeck process

The characteristic function, associated with the unconditional probability distribution $p(v_0,t_0)$ in (11.33) is also independent of time and given by

$$
G(s) = e^{-\gamma\left(\frac{\sigma s}{2}\right)^2}. \qquad (11.37)
$$

The associated moments and cumulants are

$$
\langle v^n(t)\rangle = \begin{cases} 0 & \text{for odd } n,\\ (n-1)!!(\gamma\sigma^2/2)^{n/2} & \text{otherwise.}\end{cases} \qquad (11.38)
$$

and

$$
\langle\langle v^n(t)\rangle\rangle = \begin{cases} \gamma\sigma^2/2 & \text{for } n=2,\\ 0 & \text{otherwise.}\end{cases} \qquad (11.39)
$$

The characteristic function for the conditional probability (11.34) is

$$
G(s_0,t_0;s_1,t_1) = e^{-\gamma\sigma^2\left(s_0^2+s_1^2+2s_0s_1e^{-\gamma|t_1-t_0|}\right)/4}, \qquad (11.40)
$$

The corresponding correlation functions, defined according to (11.8) are

$$\langle v^{n_1}(t_1)v^{n_0}(t_0)\rangle = \begin{cases} 0 & \text{for odd } n_0+n_1, \\ \frac{1}{2}\gamma\sigma^2 e^{-\gamma|t_1-t_0|} & \text{for } n_0=n_1=1, \\ \frac{3}{4}\gamma^2\sigma^4 e^{-\gamma|t_1-t_0|} & \text{for } n_0=1, n_1=3, \\ \frac{1}{4}\gamma^2\sigma^4\left(1+2e^{-\gamma|t_1-t_0|}\right) & \text{for } n_0=2, n_1=2, \\ \dots & . \end{cases} \tag{11.41}$$

This implies that the correlation of $v(t_1)$ and $v(t_0)$ decays exponentially. As for the Wiener process, the most compact description of the unconditional probability is given by the cumulants

$$\langle\langle v^{n_1}(t_1)v^{n_0}(t_0)\rangle\rangle = \begin{cases} \frac{1}{2}\gamma\sigma^2 e^{-\gamma|t_1-t_0|} & \text{for } n_0=n_1=1, \\ 0 & \text{otherwise for } n_0,n_1\neq 0. \end{cases} \tag{11.42}$$

The Wiener process as the integral of the Ornstein-Uhlenbeck process

We want to demonstrate now that integration of the Ornstein-Uhlenbeck process $v(t)$ yields the Wiener process. One expects the formal relationship

$$\tilde{\omega}(t) = \int_0^t ds\ v(s), \tag{11.43}$$

to hold where $\tilde{\omega}(t)$ is a Wiener process. In order to test this supposition one needs to relate the cumulants (11.9) and (11.39),(11.42) for these processes according to

$$\langle\langle\tilde{\omega}(t)\tilde{\omega}(t')\rangle\rangle = \left\langle\left\langle \int_0^t ds\ v(s)\int_0^{t'} ds'\ v(s')\right\rangle\right\rangle = \int_9^t ds\int_0^{t'} ds'\ \langle\langle v(s)v(s')\rangle\rangle\,, \tag{11.44}$$

assuming $t\geq t'$. By means of (11.42) follows

$$\begin{aligned}\langle\langle\tilde{\omega}(t)\tilde{\omega}(t')\rangle\rangle &= \frac{1}{2}\gamma\sigma^2\left[\int_0^{t'} ds'\int_0^{s'} ds\ e^{-\gamma(s'-s)} + \int_0^{t'} ds\int_0^{s} ds'\ e^{-\gamma(s-s')} + \int_{t'}^{t} ds\int_0^{t'} ds'\ e^{-\gamma(s-s')}\right] \\ &= \sigma^2 t' + \frac{\sigma^2}{2\gamma}\left[-1+e^{-\gamma t'}+e^{-\gamma t}-e^{-\gamma(t-t')}\right]. \end{aligned} \tag{11.45}$$

For times long compared to the time scale of velocity relaxation γ^{-1} one reproduces Eq. (11.24) (we don't treat explicitly the case $t\leq t'$), and for $t=t'$ Eq. (11.24), where $D=\sigma^2/2$. The relationship between the Ornstein-Uhlenbeck and Wiener processes defined through (11.43) holds for all cumulants, not just for the cumulants of second order. We only had to proof relation (11.43), since all other cumulants of both processes are simply 0. This allows one to state for

$$\omega(t) = \lim_{\gamma\to\infty}\int dt\ v(t). \tag{11.46}$$

With respect to their probability distributions the Ornstein-Uhlenbeck process $v(t)$ and the velocity of a random walker $\dot{x}(t)$ are different stochastic processes. However, in the limit $\gamma\to\infty$ and $\epsilon\to 0$ the following moment and correlation function turn out to be the same for both processes, if $D=\sigma^2/4=a^2/2\tau$

$$\langle v(t)\rangle = \langle\dot{x}(t)\rangle = 0, \tag{11.47}$$

$$\lim_{\gamma\to\infty}\langle v(t_1)v(t_0)\rangle = \lim_{\epsilon\to 0}\langle\dot{x}(t_1)\dot{x}(t_0)\rangle = 2D\delta(t_1-t_0). \tag{11.48}$$

Hence, one uses these later properties to define an idealized Markov process, the so-called *Gaussian white noise*.

F. White Noise Process

An important idealized stochastic process is the so-called "*Gaussian white noise*". This process, denoted by $\xi(t)$, is not characterized through conditional and unconditional probabilities, but through the following statistical moment and correlation function

$$\langle\xi(t)\rangle = 0, \tag{11.49}$$

$$\langle\xi(t_1)\xi(t_0)\rangle = \zeta^2\delta(t_1-t_0). \tag{11.50}$$

The attribute *Gaussian* implies that all cumulants higher than of second order are 0

$$\langle\langle \xi^{n_1}(t_1)\xi^{n_0}(t_0)\rangle\rangle = \begin{cases} \zeta^2\delta(t_1-t_0) & \text{for } n_0, n_1 = 1, \\ 0 & \text{otherwise.} \end{cases} \tag{11.51}$$

The reason why this process is termed *white* is connected with its correlation function (11.50), the Fourier transform of which is constant, i.e., entails all frequencies with equal amplitude just as white radiation. The importance of the process $\xi(t)$ stems from the fact that many other stochastic processes are described through stochastic differential equations with a (white) noise term $\xi(t)$. We will show this for the Wiener process below and for the Ornstein-Uhlenbeck processes later in the notes.

As hinted by the examples in this section we can show that the integral of Gaussian white noise happens to be a Wiener process. We prove this in the same fashion as above by deriving the cumulants for $\int dt\ \xi(t)$. Again the task is simplified by the fact that only one cumulant is non-zero, namely,

$$\begin{aligned} \left\langle\left\langle \int_0^{t_1} ds_1\ \xi(s_1) \int_0^{t_0} ds_0\ \xi(s_0) \right\rangle\right\rangle &= \int_0^{t_1} ds_1 \int_0^{t_0} ds_0\ \langle\langle \xi(t_1)\xi(t_0)\rangle\rangle \\ &= \int_0^{t_1} ds_1 \int_0^{t_0} ds_0\ \zeta^2\delta(t_1-t_0) \\ &= \zeta^2 \min(t_1, t_0). \end{aligned} \tag{11.52}$$

We demonstrated, thereby, the important relationship between white noise $\xi(t)$ and the Wiener process $\omega(t)$

$$\omega(t) = \int_0^t ds\ \xi(s), \tag{11.53}$$

for $2D = \zeta^2 = 1$.

XII. STOCHASTIC DIFFERENTIAL EQUATIONS

XIII. ITO AND STRATONOVICH CALCULUS

XIV. EINSTEIN DIFFUSION EQUATION

XV. SMOLUCHOWSKI DIFFUSION EQUATION

XVI. ADJOINT SMOLUCHOWSKI DIFFUSION EQUATION

Appendix A: Stirling approximation and asymptotic binomial distribution

In order to prove Gauß asymptotic approximation (11.32) of the binomial distribution we need Stirling formula

$$n! = \sqrt{2\pi n}\left(\frac{n}{e}\right)^n \left(1 + O\left(\frac{1}{n}\right)\right). \tag{A1}$$

We set forth to prove Eq. (11.32), i.e.,

$$\sqrt{\frac{n}{2}}2^{-n}\binom{n}{\frac{n}{2}+x\sqrt{\frac{n}{2}}} = \frac{1}{\sqrt{\pi}}e^{-x^2}\left(1 + O\left(\frac{1}{n}\right)\right). \tag{A2}$$

Applying the natural logarithm on both sides of this equation we obtain

$$\ln\left[\sqrt{\frac{n}{2}}2^{-n}\binom{n}{\frac{n}{2}+x\sqrt{\frac{n}{2}}}\right] = -\frac{1}{2}\ln\pi - x^2 + O\left(\frac{1}{n}\right). \tag{A3}$$

We will prove this equation by transforming the left hand side step by step. First, we utilize the formula $\binom{n}{k} = \frac{n!}{k!(n-k)!}$ for binomial coefficients.

$$
\begin{aligned}
&\ln\left[\sqrt{\frac{n}{2}}2^{-n}\binom{n}{\frac{n}{2}+x\sqrt{\frac{n}{2}}}\right] \\
&= \frac{1}{2}(\ln n - \ln 2) - n\ln 2 + \ln\left(\frac{n!}{\left(\frac{n}{2}+x\sqrt{\frac{n}{2}}\right)!\left(\frac{n}{2}-x\sqrt{\frac{n}{2}}\right)!}\right) \\
&= \frac{1}{2}\ln n - \left(n+\frac{1}{2}\right)\ln 2 + \ln n! - \ln\left[\left(\frac{n}{2}+x\sqrt{\frac{n}{2}}\right)!\right] - \ln\left[\left(\frac{n}{2}-x\sqrt{\frac{n}{2}}\right)!\right]. \qquad \text{(A4)}
\end{aligned}
$$

Applying Stirling formula (A1) we derive furthermore

$$
\begin{aligned}
&= \frac{1}{2}\ln n - \left(n+\frac{1}{2}\right)\ln 2 + \frac{1}{2}\ln(2\pi) + \left(n+\frac{1}{2}\right)\ln n - n + \\
&\quad -\frac{1}{2}\ln(2\pi) - \left(\frac{n}{2}+x\sqrt{\frac{n}{2}}+\frac{1}{2}\right)\ln\left(\frac{n}{2}+x\sqrt{\frac{n}{2}}\right) + \frac{n}{2} + x\sqrt{\frac{n}{2}} + \\
&\quad -\frac{1}{2}\ln(2\pi) - \left(\frac{n}{2}-x\sqrt{\frac{n}{2}}+\frac{1}{2}\right)\ln\left(\frac{n}{2}-x\sqrt{\frac{n}{2}}\right) + \frac{n}{2} - x\sqrt{\frac{n}{2}} \\
&= \frac{1}{2}\ln n - \left(n+\frac{1}{2}\right)\ln 2 - \frac{1}{2}\ln(2\pi) + \left(n+\frac{1}{2}\right)\ln n + \\
&\quad -\left(\frac{n}{2}+x\sqrt{\frac{n}{2}}+\frac{1}{2}\right)\ln\left(\frac{n}{2}+x\sqrt{\frac{n}{2}}\right) + \\
&\quad -\left(\frac{n}{2}-x\sqrt{\frac{n}{2}}+\frac{1}{2}\right)\ln\left(\frac{n}{2}-x\sqrt{\frac{n}{2}}\right) \\
&= (n+1)\ln n - \left(n+\frac{1}{2}\right)\ln 2 - \frac{1}{2}\ln(2\pi) + \\
&\quad -\left(\frac{n}{2}+x\sqrt{\frac{n}{2}}+\frac{1}{2}\right)\left(\ln\frac{n}{2} + \ln\left(1+x\sqrt{\frac{2}{n}}\right)\right) + \\
&\quad -\left(\frac{n}{2}-x\sqrt{\frac{n}{2}}+\frac{1}{2}\right)\left(\ln\frac{n}{2} + \ln\left(1-x\sqrt{\frac{2}{n}}\right)\right).
\end{aligned}
$$

Performing a Taylor expansion of $\ln(1 \pm z)$ with respect to z we obtain

$$
\begin{aligned}
&= (n+1)(\ln n - \ln 2) - \frac{1}{2}\ln\pi + \\
&\quad -\left(\frac{n}{2}+x\sqrt{\frac{n}{2}}+\frac{1}{2}\right)\left(\ln\frac{n}{2} + x\sqrt{\frac{2}{n}} - \frac{x^2}{n} + \frac{x^3}{3}\sqrt{\frac{8}{n^3}} + O\left(\frac{1}{n^2}\right)\right) + \\
&\quad -\left(\frac{n}{2}-x\sqrt{\frac{n}{2}}+\frac{1}{2}\right)\left(\ln\frac{n}{2} - x\sqrt{\frac{2}{n}} - \frac{x^2}{n} - \frac{x^3}{3}\sqrt{\frac{8}{n^3}} + O\left(\frac{1}{n^2}\right)\right),
\end{aligned}
$$

and expanding the products up to order $O(1/n)$ we acquire the result, the right hand side of Eq. (A3),

$$
\begin{aligned}
&= (n+1)(\ln n - \ln 2) - \frac{1}{2}\ln\pi + \\
&\quad -\frac{n}{2}\ln\frac{n}{2} - x\sqrt{\frac{n}{2}} + \frac{x^2}{2} - \frac{x^3}{3}\sqrt{\frac{2}{n}} - x\sqrt{\frac{n}{2}}\ln\frac{n}{2} - x^2 + \frac{x^3}{\sqrt{2n}} - \frac{1}{2}\ln\frac{n}{2} - \frac{x}{\sqrt{2n}} + O\left(\frac{1}{n}\right) + \\
&\quad -\frac{n}{2}\ln\frac{n}{2} + x\sqrt{\frac{n}{2}} + \frac{x^2}{2} + \frac{x^3}{3}\sqrt{\frac{2}{n}} + x\sqrt{\frac{n}{2}}\ln\frac{n}{2} - x^2 - \frac{x^3}{\sqrt{2n}} - \frac{1}{2}\ln\frac{n}{2} + \frac{x}{\sqrt{2n}} + O\left(\frac{1}{n}\right) \\
&= (n+1)\ln\frac{n}{2} - \frac{1}{2}\ln\pi - n\ln\frac{n}{2} - x^2 - \ln\frac{n}{2} + O\left(\frac{1}{n}\right) \\
&= -\frac{1}{2}\ln\pi - x^2 + O\left(\frac{1}{n}\right).
\end{aligned}
$$

AUTHOR DECLARATIONS

Conflicts of interest

None declared.

Data availability

The data that support the findings of this study are available from the corresponding author upon reasonable request.

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